

## CALCASIEU PASS, LA, USA

WMO: 997337

Lat: 29.7641N

Lon: 93.3428W

Elev: 3

StdP: 101.29

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |            | Humidification DP/MCDB and HR |            |            |            |            |            | Coldest Month WS/MCDB |            |             |            | MCWS/PCWD to 99.6% DB |           | WSF          |
|---------------|------------|------------|-------------------------------|------------|------------|------------|------------|------------|-----------------------|------------|-------------|------------|-----------------------|-----------|--------------|
|               |            |            | 99.6%                         |            |            | 99%        |            |            | 0.4%                  |            | 1%          |            |                       |           |              |
|               | 99.6%      | 99%        | DP                            | HR         | MCDB       | DP         | HR         | MCDB       | WS                    | MCDB       | WS          | MCDB       | MCWS                  | PCWD      |              |
| (a)<br>1      | (b)<br>0.1 | (c)<br>2.5 | (d)<br>N/A                    | (e)<br>N/A | (f)<br>N/A | (g)<br>N/A | (h)<br>N/A | (i)<br>N/A | (j)<br>12.1           | (k)<br>8.6 | (l)<br>11.3 | (m)<br>8.7 | (n)<br>6.0            | (o)<br>30 | (p)<br>0.518 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |            |             |            |             |            | Evaporation WB/MCDB |            |            |            |            |            | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|------------|-------------|------------|-------------|------------|---------------------|------------|------------|------------|------------|------------|----------------------|------------|
|               |                        | 0.4%            |            | 1%          |            | 2%          |            | 0.4%                |            | 1%         |            | 2%         |            |                      |            |
|               |                        | DB              | MCWB       | DB          | MCWB       | DB          | MCWB       | WB                  | MCDB       | WB         | MCDB       | WB         | MCDB       | MCWS                 | PCWD       |
| (a)<br>8      | (b)<br>3.7             | (c)<br>31.8     | (d)<br>N/A | (e)<br>31.0 | (f)<br>N/A | (g)<br>30.6 | (h)<br>N/A | (i)<br>N/A          | (j)<br>N/A | (k)<br>N/A | (l)<br>N/A | (m)<br>N/A | (n)<br>N/A | (o)<br>4.3           | (p)<br>220 |

| Dehumidification DP/MCDB and HR |            |            |            |            |            |            |            |            | Enthalpy/MCDB |            |            |            |            |            | Extreme<br>Max WB |
|---------------------------------|------------|------------|------------|------------|------------|------------|------------|------------|---------------|------------|------------|------------|------------|------------|-------------------|
| 0.4%                            |            |            | 1%         |            |            | 2%         |            |            | 0.4%          |            | 1%         |            | 2%         |            |                   |
| DP                              | HR         | MCDB       | DP         | HR         | MCDB       | DP         | HR         | MCDB       | Enth          | MCDB       | Enth       | MCDB       | Enth       | MCDB       |                   |
| (a)<br>N/A                      | (b)<br>N/A | (c)<br>N/A | (d)<br>N/A | (e)<br>N/A | (f)<br>N/A | (g)<br>N/A | (h)<br>N/A | (i)<br>N/A | (j)<br>N/A    | (k)<br>N/A | (l)<br>N/A | (m)<br>N/A | (n)<br>N/A | (o)<br>N/A | (p)<br>N/A        |

## Extreme Annual Design Conditions

| Extreme Annual WS |          |         | Extreme Annual Temperature |         |                    |         | n-Year Return Period Values of Extreme Temperature |         |            |         |            |         |            |         |  |  |
|-------------------|----------|---------|----------------------------|---------|--------------------|---------|----------------------------------------------------|---------|------------|---------|------------|---------|------------|---------|--|--|
|                   |          |         | Mean                       |         | Standard Deviation |         | n=5 years                                          |         | n=10 years |         | n=20 years |         | n=50 years |         |  |  |
| 1%                | 2.5%     | 5%      | Min                        | Max     | Min                | Max     | Min                                                | Max     | Min        | Max     | Min        | Max     | Min        | Max     |  |  |
| (a) 11.6          | (b) 10.6 | (c) 9.3 | (e) N/A                    | (f) N/A | (g) N/A            | (h) N/A | (i) N/A                                            | (j) N/A | (k) N/A    | (l) N/A | (m) N/A    | (n) N/A | (o) N/A    | (p) N/A |  |  |
|                   |          |         | DB                         | N/A     | N/A                | N/A     | N/A                                                | N/A     | N/A        | N/A     | N/A        | N/A     | N/A        | N/A     |  |  |
|                   |          |         | WB                         | N/A     | N/A                | N/A     | N/A                                                | N/A     | N/A        | N/A     | N/A        | N/A     | N/A        | N/A     |  |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 20.6   | 10.8  | 12.7  | 16.6  | 20.3  | 24.3  | 27.7  | 28.2  | 28.7  | 26.4  | 21.9  | 15.6  | 13.1 |  |  |
|                                                                        | DBStd       | 7.15   | 4.56  | 4.63  | 4.07  | 3.31  | 2.41  | 1.45  | 1.26  | 1.41  | 1.78  | 3.64  | 4.26  | 4.28 |  |  |
|                                                                        | HDD10.0     | 108    | 46    | 27    | 6     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 9     | 21   |  |  |
|                                                                        | HDD18.3     | 773    | 233   | 161   | 79    | 20    | 2     | 0     | 0     | 0     | 0     | 12    | 98    | 168  |  |  |
|                                                                        | CDD10.0     | 3969   | 72    | 102   | 210   | 309   | 444   | 532   | 563   | 581   | 493   | 371   | 177   | 115  |  |  |
|                                                                        | CDD18.3     | 1592   | 1     | 3     | 25    | 78    | 188   | 282   | 304   | 323   | 243   | 124   | 16    | 4    |  |  |
|                                                                        | CDH23.3     | 16242  | 0     | 1     | 7     | 145   | 1319  | 3371  | 3821  | 4216  | 2499  | 861   | 3     | 0    |  |  |
| (14) Wind                                                              | WSAvg       | 5.0    | 4.9   | 5.0   | 5.1   | 5.8   | 5.9   | 5.2   | 4.7   | 4.5   | 4.3   | 4.8   | 5.0   | 5.0  |  |  |
|                                                                        | PrecAvg     | 1554   | 150   | 96    | 102   | 117   | 105   | 169   | 156   | 119   | 132   | 131   | 122   | 123  |  |  |
|                                                                        | PrecMax     | 2121   | 344   | 218   | 256   | 230   | 322   | 255   | 243   | 253   | 327   | 529   | 356   | 252  |  |  |
|                                                                        | PrecMin     | 1061   | 36    | 18    | 46    | 12    | 0     | 36    | 77    | 13    | 19    | 38    | 16    | 60   |  |  |
|                                                                        | PrecStd     | 294    | 86    | 63    | 56    | 61    | 82    | 62    | 47    | 66    | 79    | 120   | 80    | 55   |  |  |
|                                                                        | DB          | 19.7   | 22.3  | 24.2  | 26.3  | 29.3  | 32.8  | 31.9  | 34.2  | 32.1  | 29.7  | 24.1  | 21.9  |      |  |  |
|                                                                        | MCWB        | N/A    | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |      |  |  |
| (20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 2% DB       | 18.6   | 20.8  | 23.3  | 25.2  | 28.1  | 31.0  | 30.8  | 32.1  | 30.6  | 28.6  | 23.0  | 20.8  |      |  |  |
|                                                                        | MCWB        | N/A    | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |      |  |  |
|                                                                        | 5% DB       | 17.9   | 19.1  | 22.6  | 24.7  | 27.5  | 30.2  | 30.3  | 31.3  | 30.0  | 27.9  | 22.2  | 20.0  |      |  |  |
|                                                                        | MCWB        | N/A    | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |      |  |  |
|                                                                        | 10% DB      | 17.1   | 18.3  | 21.7  | 24.2  | 27.0  | 29.8  | 30.1  | 30.8  | 29.4  | 27.1  | 21.3  | 19.1  |      |  |  |
|                                                                        | MCWB        | N/A    | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |      |  |  |
|                                                                        | MCDB        | N/A    | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |      |  |  |
| (28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4% WB     | N/A    | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |      |  |  |
|                                                                        | MCDB        | N/A    | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |      |  |  |
|                                                                        | 2% WB       | N/A    | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |      |  |  |
|                                                                        | MCDB        | N/A    | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |      |  |  |
|                                                                        | 5% WB       | N/A    | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |      |  |  |
|                                                                        | MCDB        | N/A    | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |      |  |  |
|                                                                        | 10% WB      | N/A    | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |      |  |  |
| (35) Mean Daily Temperature Range                                      | MDBR        | 5.9    | 5.2   | 4.9   | 4.8   | 3.6   | 3.3   | 3.5   | 3.7   | 4.8   | 6.1   | 6.2   | 6.0   |      |  |  |
|                                                                        | 5% DB       | 5.1    | 4.5   | 4.4   | 4.3   | 3.8   | 4.3   | 3.7   | 4.5   | 5.3   | 5.2   | 5.3   | 5.4   |      |  |  |
|                                                                        | MCWBR       | N/A    | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |      |  |  |
|                                                                        | 5% WB       | N/A    | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |      |  |  |
|                                                                        | MCWBR       | N/A    | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |      |  |  |
|                                                                        | taub        | 0.347  | 0.362 | 0.379 | 0.432 | 0.471 | 0.484 | 0.494 | 0.479 | 0.455 | 0.386 | 0.371 | 0.355 |      |  |  |
|                                                                        | taud        | 2.518  | 2.487 | 2.430 | 2.279 | 2.204 | 2.217 | 2.221 | 2.267 | 2.298 | 2.476 | 2.456 | 2.495 |      |  |  |
| (42) Clear-Sky Solar Irradiance                                        | Ebn at Noon | 887    | 904   | 907   | 866   | 830   | 814   | 804   | 814   | 823   | 866   | 851   | 858   |      |  |  |
|                                                                        | Edh at Noon | 89     | 100   | 112   | 135   | 146   | 144   | 143   | 135   | 127   | 100   | 94    | 87    |      |  |  |
|                                                                        | RadAvg      | 2.91   | 3.35  | 4.65  | 5.62  | 6.31  | 6.36  | 6.01  | 5.67  | 5.05  | 4.49  | 3.38  | 2.60  |      |  |  |
| (45) All-Sky Solar Radiation                                           | RadStd      | 0.30   | 0.54  | 0.43  | 0.43  | 0.40  | 0.46  | 0.47  | 0.48  | 0.53  | 0.54  | 0.32  | 0.28  |      |  |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (28 neighbors) |  | +0.33 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +107    | +82     |

Nomenclature: See separate page